

Multiple Choice (10 marks)

1. A refracting telescope consists of two converging lenses separated by 100 cm. The eye-piece lens has a focal length of 20 cm. The angular magnification of the telescope is
 - (a) 4
 - (b) 5
 - (c) 6
 - (d) 20
2. If the absolute temperature of a blackbody is increased by a factor of 3, the energy radiated per second per unit area does which of the following?
 - (a) Decreases by a factor of 81.
 - (b) Decreases by a factor of 9.
 - (c) Increases by a factor of 9.
 - (d) Increases by a factor of 81.
3. According to the BCS theory, the attraction between Cooper pairs in a superconductor is due to
 - (a) the weak nuclear force
 - (b) the strong nuclear force
 - (c) vacuum polarization
 - (d) interactions with the ionic lattice
4. The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately
 - (a) $0.1 \text{ GeV}/c^2$
 - (b) $0.2 \text{ GeV}/c^2$
 - (c) $0.5 \text{ GeV}/c^2$
 - (d) $1.0 \text{ GeV}/c^2$
5. The negative muon, μ^- , has properties most similar to which of the following?
 6. Electron
 7. Meson
 8. Photon
 9. Boson

True / False (10 marks)

Answer True or False

6. True or False: An atom with filled $n = 1$ and $n = 2$ energy levels contains a total of 10 electrons.
7. True or False: A single-electron atom with the electron in the $l = 2$ state has only three allowed values for the magnetic quantum number m_l .
8. True or False: In the lab frame, a particle moving at $v=0.60 c$ will travel 450 m before decaying.
9. True or False: A refracting telescope with a focal length of 20 cm for the eye-piece has an angular magnification of 6.
10. True or False: A grating spectrometer can have a resolving power of 5,000 when resolving wavelengths that are only 2 nm apart.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the mechanism by which characteristic X rays are produced when high-energy electrons bombard a metal target, emphasizing the role of electron transitions in inner shell vacancies.
12. Discuss the characteristics of fermions and bosons, focusing on their wave functions and the implications of the Pauli exclusion principle in quantum mechanics.

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